Maths Homework Template

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Short Math Guide can see `http://tug.ctan.org/info/short-math-guide`, and it's Chinese version here: `https://github.com/hoganbin/short-math-guide`.

1. Answer to question 1:

   Use inline equations for simple math $1+1=2$, and centered equations for more involved or important equations:

   $$a^2+b^2=c^2. \tag{1}$$

   $$\frac{\mathrm{d}y}{\mathrm{d}x}=\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} \tag{2}$$

   It is recommended to write scalars without boldface like $E=\hbar\omega$, vectors in boldface, and matrices in sans-serif:

   $$\boldsymbol{p}=\hbar\boldsymbol{k} \tag{3}$$

   $$\frac{\|\mathsf{A}\boldsymbol{x}\|_\infty}{\|\boldsymbol{x}\|_\infty}=\frac{\max\limits_i\left|\sum\limits_{j=1}^{n}\mathsf{A}_{ij}\mathsf{x}_{j1}\right|}{\max\limits_i|\mathsf{x}_{i1}|} \tag{4}$$

   ¶ An example of a matrix:

   $$\mathsf{M}=\begin{bmatrix}\dfrac{1+\sqrt{3}}{2} & \mathrm{e}^{-i\omega t}\\ -\mathrm{e}^{-i\omega t} & \dfrac{1-\sqrt{3}}{2}\end{bmatrix} \tag{5}$$

   ¶ An example of intergals:

   $$\iiint\nabla\cdot\boldsymbol{\varPhi}\,\mathrm{d}V=\oiint\boldsymbol{\varPhi}\,\mathrm{d}\boldsymbol{S} \tag{6}$$

   ¶ An example of roots:

   $$r=\sqrt[3]{\frac{3}{4\pi N}} \tag{7}$$

2. Answer to question 2:

$$\begin{aligned}I(x)&=\int_{-\frac{b}{2}}^{\frac{b}{2}}\mathrm{d}I(x)\\&=4\alpha\frac{I_0}{b}\int_{-\frac{b}{2}}^{\frac{b}{2}}\cos^2\left(\frac{\pi d}{\lambda D}\left(x+\frac{D}{R}\xi\right)\right)\mathrm{d}\xi\\&=4\alpha I_0\left(1+\frac{1}{u}\sin u\cos\left(\frac{2\pi d}{\lambda D}x\right)\right),\quad u=\frac{\pi db}{\lambda R}\end{aligned}$$

3. Answer to question 3:

$$Y_n(y) = \begin{cases} 0 \qquad , & n = 2k \\ \dfrac{8a^2}{\pi^3 n^3}\left(1 - \dfrac{\cosh\dfrac{n\pi y}{a}}{\cosh\dfrac{n\pi b}{2a}}\right), & n = 2k+1 \end{cases}$$

4. Answer to question 4:

$$\begin{aligned} u(x,t) = {} & \frac{6Al(-1)^m}{ES(2m+1)^2\pi^2}\sin\frac{2m+1}{2l}\pi x\sin\frac{2m+1}{2l}\pi at \\ & + \frac{2A(-1)^{m+1}}{ES(2m+1)\pi}\left(x\cos\frac{2m+1}{2l}\pi x\sin\frac{2m+1}{2l}\pi at + at\sin\frac{2m+1}{2l}\pi x\cos\frac{2m+1}{2l}\pi at\right) \\ & + \frac{8Al(2m+1)}{ES\pi^2}\sum_{k=0,\, k\neq m}^{\infty}\frac{(-1)^k}{(2k+1)\left[(2m+1)^2-(2k+1)^2\right]}\sin\frac{2k+1}{2l}\pi x\sin\frac{2k+1}{2l}\pi at \end{aligned}$$

5. Answer to question 5:

$$\begin{aligned} u(x,t) &= \sum_{n=0}^{\infty} X_n(x)Y_n(y) \\ &= \frac{8a^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\left(1 - \frac{\cosh\dfrac{2k+1}{a}\pi y}{\cosh\dfrac{2k+1}{2a}\pi b}\right)\sin\frac{2k+1}{a}\pi x \end{aligned}$$

6. Answer to question 6:

$$\begin{aligned} \mathscr{L}\left\{\frac{1-\cos\omega t}{t}\right\} &= \int_p^{\infty}\left(\frac{1}{p} - \frac{1}{2(p+\mathrm{i}\omega)} - \frac{1}{2(p-\mathrm{i}\omega)}\right)\mathrm{d}t \\ &= \ln p - \frac{1}{2}\ln(p+\mathrm{i}\omega) - \frac{1}{2}\ln(p-\mathrm{i}\omega)\Big|_p^{\infty} \\ &= \frac{1}{2}\ln\frac{p^2}{p^2+\omega^2}\Big|_p^{\infty} \\ &= \frac{1}{2}\ln\frac{p^2+\omega^2}{p^2} \end{aligned}$$

7. Answer to question 7:

$$\begin{aligned} [q,p]_{Q,P} &= \frac{\partial q}{\partial Q}\cdot\frac{\partial p}{\partial P} - \frac{\partial q}{\partial P}\cdot\frac{\partial p}{\partial Q} \\ &= \left(-\frac{\sqrt{1-P^2\mathrm{e}^{2Q}}}{\mathrm{e}^Q} - \frac{P^2\mathrm{e}^Q}{\sqrt{1-P^2\mathrm{e}^{2Q}}}\right)\cdot\left(\frac{-\mathrm{e}^Q}{\sqrt{1-P^2\mathrm{e}^{2Q}}}\right) - \left(\frac{-P\mathrm{e}^Q}{\sqrt{1-P^2\mathrm{e}^{2Q}}}\right)\cdot\left(\frac{-P\mathrm{e}^Q}{\sqrt{1-P^2\mathrm{e}^{2Q}}}\right) \\ &= 1 + \frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}} - \frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}} \\ &= 1 \end{aligned}$$

8. Answer to question 8:

$$\frac{4\pi R_\odot^2\cdot\sigma T_\odot^4}{4\pi d^2} = 1352.83\,\mathrm{W/m^2}$$
$$T_\odot = 5763.43\,\mathrm{K}$$
$$\lambda_\odot = \frac{b}{T_\odot} = 5027.86\,\text{Å}$$